

# Companion to Bird & de Moor, Algebra of Programming

## §6. Recursive Programs

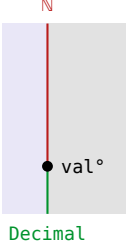
### §6.1. Digits of a number

(6.1a)

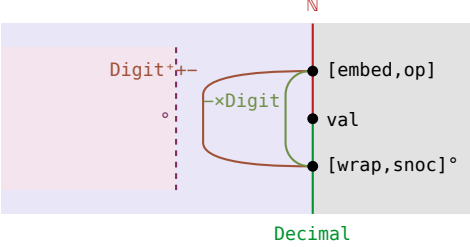
$\text{val}^\circ = \text{embed}^\circ \text{ wrap } \cup \text{ op}^\circ \text{ } \neg\text{Digit}(\text{val}^\circ) \text{ snoc}$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

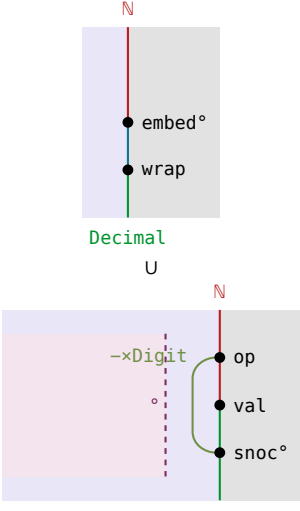
the converse of  $\text{val}$ , which reads digits as a number



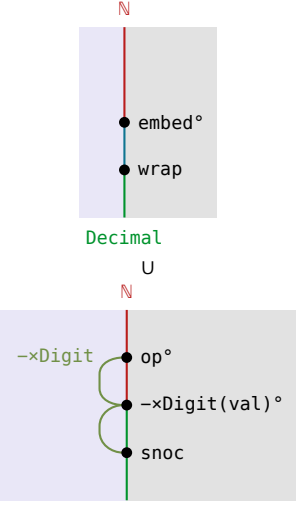
(a)



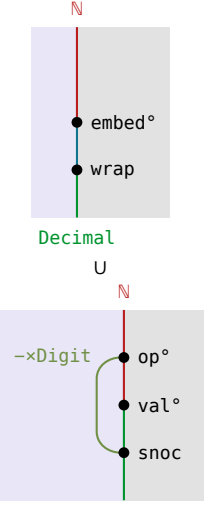
(b)



(c)



(d)



(e)

|     |                                                                                                                         |                                                                                                                                                                            |
|-----|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{val}^\circ$                                                                                                      | $\text{val}$ undoes $\text{wrap}$ or $\text{snoc}$ , reads the front with $\text{val}$ and combines by $\text{embed}$ or $\text{op}$ , so its converse runs that backwards |
| (b) | $([\text{wrap}, \text{snoc}]^\circ \text{ Digit}^{+--} (\neg\text{Digit}(\text{val}))) [\text{embed}, \text{op}]^\circ$ | undoing $\text{wrap}$ or $\text{snoc}$ first gives two cases: a decimal and its last digit read by $\text{val}$ then $\text{op}$ , or one digit read by $\text{embed}$     |
| (c) | $(\text{wrap}^\circ \text{ embed})^\circ \cup (\text{snoc}^\circ \neg\text{Digit}(\text{val}) \text{ op})^\circ$        | the converse of each case is its steps undone in reverse: undo $\text{op}$ , undo reading the front, then $\text{snoc}$ ; or undo $\text{embed}$ , then $\text{wrap}$      |
| (d) | $\text{embed}^\circ \text{ wrap } \cup \text{ op}^\circ \neg\text{Digit}(\text{val})^\circ \text{ snoc}$                | undoing the read of the front while keeping the last digit is $\text{val}^\circ$ on the front alone                                                                        |
| (e) | $\text{embed}^\circ \text{ wrap } \cup \text{ op}^\circ \neg\text{Digit}(\text{val}^\circ) \text{ snoc}$                |                                                                                                                                                                            |

(6.1b)

|                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|
| $\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$                                       |
| $\text{op}^\circ$ splits a number into its quotient and remainder by 10                                      |
| $(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$                                       |
| $\text{op}^\circ$ gives a pair with a nonzero first component exactly at the numbers with two or more digits |
| $(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$                                             |
| $\text{embed}^\circ$ gives a digit exactly at the one-digit numbers                                          |

## §6.2. Least fixed points

(6.2a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$ .

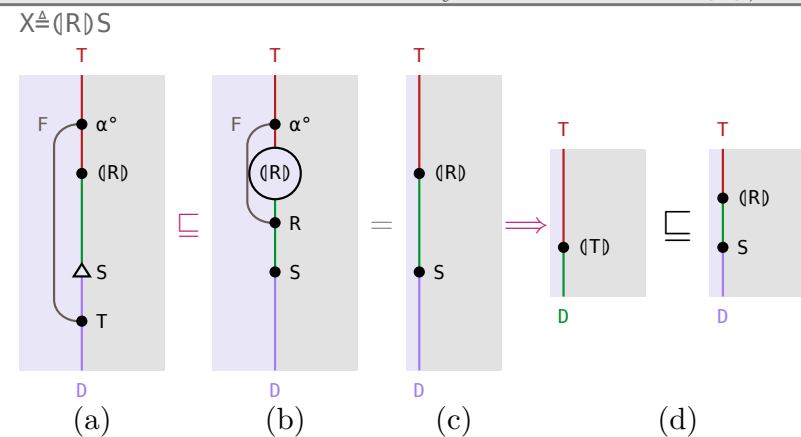
$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

(6.2b)

| the law                                                                                    | what it says                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 6.3 and every chapter after it uses                                      |
| $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |
| $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$<br>(6.2)   | a relation $X$ that contains one unfolding of the fold's recursion at $X$ contains the fold                                                                                     |
| $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$<br>(6.3)   | a relation $X$ contained in one unfolding of the fold's recursion at $X$ is contained in the fold                                                                               |

$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S \quad (6.2c)$$

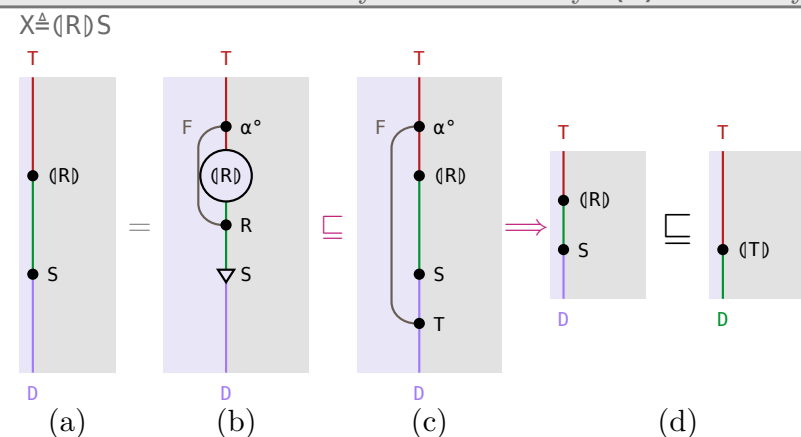
if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$



|     |                                                                                                                                       |                                                                                                                                                                                                    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\alpha^\circ F(\langle R \rangle) F(S) T$                                                                                            | $F(S) T \sqsubseteq R S$                                                                                                                                                                           |
| (b) | $\alpha^\circ F(\langle R \rangle) R S$                                                                                               | $F(\langle R \rangle) R = \alpha(\langle R \rangle)$ , $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce                                                                                             |
| (c) | $\langle R \rangle S$                                                                                                                 | $\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-)T$ , and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$ |
| (d) | $\alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$ |                                                                                                                                                                                                    |

$$R S \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle \quad (6.2d)$$

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



|     |                                                                                                                                       |                                                                                                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle R \rangle S$                                                                                                                 | $\alpha(\langle R \rangle) = F(\langle R \rangle) R$ , $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce                                                                                                 |
| (b) | $\alpha^\circ F(\langle R \rangle) R S$                                                                                               | $R S \sqsubseteq F(S) T$                                                                                                                                                                               |
| (c) | $\alpha^\circ F(\langle R \rangle S) T$                                                                                               | $\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-)T$ , and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$ |
| (d) | $\langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$ |                                                                                                                                                                                                        |

## §6.3. Hylomorphisms

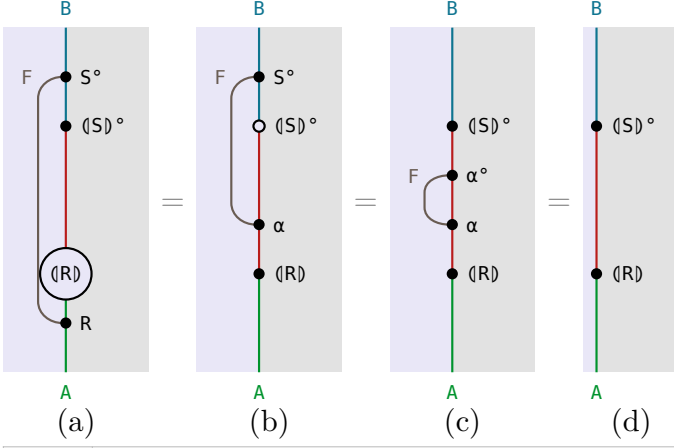
$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R) \quad (6.3a)$$

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R = (\mathbb{S})^\circ (\mathbb{R})$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term  $S^\circ$  represents the decomposition stage,  $F((\mathbb{S})^\circ (\mathbb{R}))$  the stage of solving the subproblems recursively, and  $R$  the recombination stage;  $R : FA \rightarrow A$ ,  $S : FB \rightarrow B$ ,  $\alpha : FT \rightarrow T$  initial

the body at  $(\mathbb{S})^\circ (\mathbb{R})$

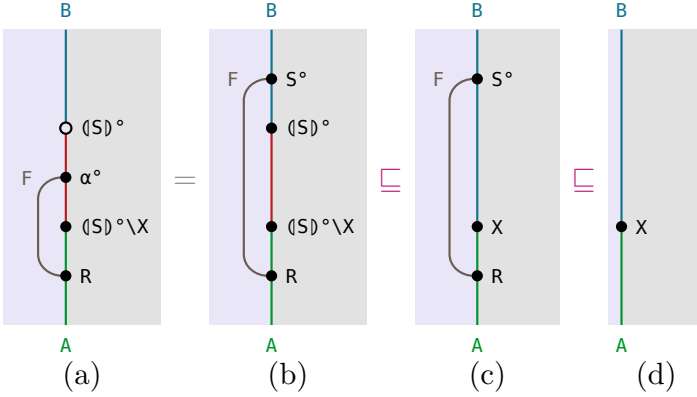


|     |                                                       |                                                                                                          |
|-----|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| (a) | $S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$     | $F((\mathbb{R})) R = \alpha(\mathbb{R}) \triangleq \text{reduce}$                                        |
| (b) | $S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$   | $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$ |
| (c) | $(\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$ | $\alpha$ iso                                                                                             |
| (d) | $(\mathbb{S})^\circ (\mathbb{R})$                     |                                                                                                          |

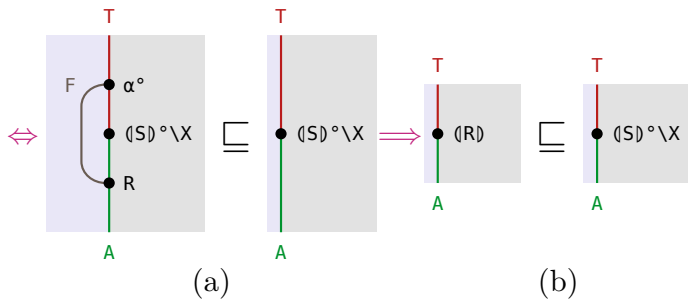
$$S^\circ F(X) R \sqsubseteq X \implies (\mathbb{S})^\circ (\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism  $(\mathbb{S})^\circ (\mathbb{R})$  is included in  $X$  if  $X$  satisfies the associated recursion inequation

$Y := (\mathbb{S})^\circ \setminus X$



|     |                                                                       |                                                                                       |
|-----|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| (a) | $(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$ | $(\mathbb{S})^\circ \alpha^\circ = S^\circ F((\mathbb{S})^\circ)$                     |
| (b) | $S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$    | $(\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \text{ — adj-all}$ |
| (c) | $S^\circ F(X) R$                                                      | $S^\circ F(X) R \sqsubseteq X$                                                        |
| (d) | $X$                                                                   | $S \cdot \neg S \text{ — adj-all}$                                                    |

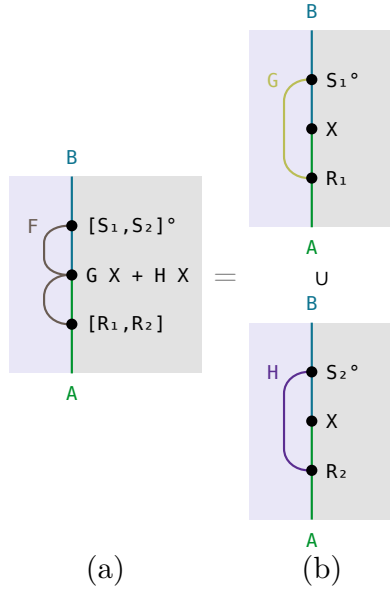


|     |                                                                                                                                     |                                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| (a) | $S^\circ F(X) R \sqsubseteq X \implies \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \sqsubseteq (\mathbb{S})^\circ \setminus X$ | $(\mathbb{R}) = (\mu X : \alpha^\circ F(X) R) \triangleq \text{reduce, (6.2b), adj-all}$ |
| (b) | $S^\circ F(X) R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$                                     |                                                                                          |

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \Rightarrow ([S_1, S_2])^\circ ([R_1, R_2]) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

the body of Theorem 6.2 at  $F(X)=G(X)+H(X)$

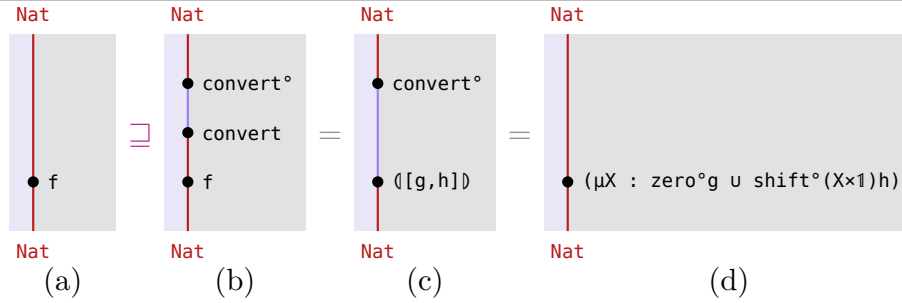


|     |                                              |                                                                                                                                                                                            |
|-----|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$  | $[S, T] = (S+T)\nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$ ; the + stands for the codiagonals $\nabla^\circ, \nabla$ , which are not drawn |
| (b) | $S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$ | $\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct                                                                                                                                         |

## §6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once  $f$  sends **zero** to  $g$  and turns **shift** into  $h$ , the recursion that halves the argument at each step computes  $f$



|     |                                                                        |                                                                |
|-----|------------------------------------------------------------------------|----------------------------------------------------------------|
| (a) | $f$                                                                    | convert simple                                                 |
| (b) | $\text{convert}^\circ \text{ convert } f$                              | fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$ |
| (c) | $\text{convert}^\circ ([g, h])$                                        | Corollary 6.1, (6.3b)                                          |
| (d) | $(\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h)$ |                                                                |

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes  $a$  to the power  $b$

$\text{zero } \text{exp}(a) = \text{one}$

$a$  to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

$a$  to the power  $2n+d$  is  $\text{op}(a)$  of  $a$  to the power  $n$  and  $d$

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves  $a$  at each step computes the remainder of  $a$  by  $b$

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \text{ mod } b$  is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

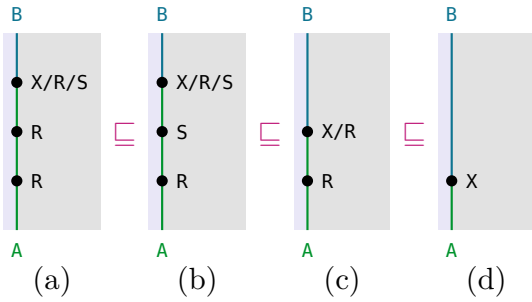
$(2a+d) \text{ mod } b$  is  $\text{op}(b)$  of  $a \text{ mod } b$  and  $d$

## §6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \ R \sqsubseteq S \ R \implies \text{inductive}(R)$$

(6.5a)

$R$  is inductive when  $RR \sqsubseteq SR$  for an inductive  $S$



|     |                   |                     |
|-----|-------------------|---------------------|
| (a) | $(X / R / S) R R$ | $RR \sqsubseteq SR$ |
| (b) | $(X / R / S) S R$ | division by $S$     |
| (c) | $(X / R) R$       | division by $R$     |
| (d) | $X$               |                     |

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of  $F+G$  is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of  $F \times G$  is a member of either component

$$\text{inductive}(S \text{ member}) \wedge X = S F(X) R \wedge Y = S F(Y) R \implies X = Y$$

(6.5c)

when  $S \text{ member}$  is inductive, two solutions  $X, Y$  of  $X = SF(X)R$  are equal

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S F(X) R \sqsubseteq X \implies \text{Entire}(X)$$

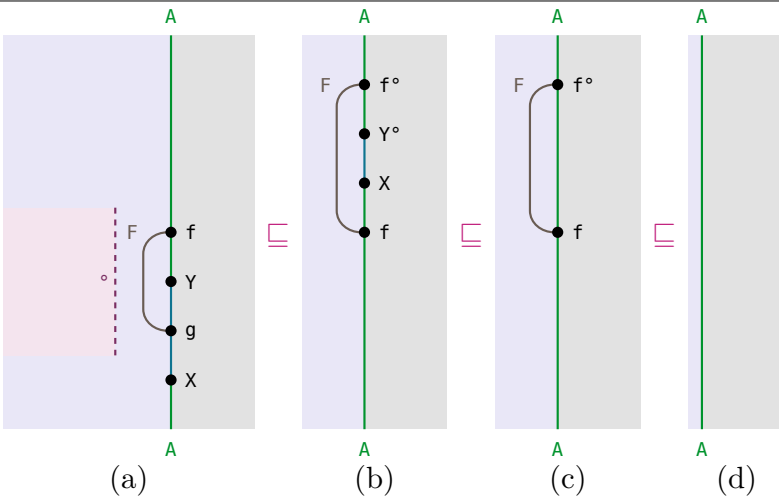
(6.5d)

when  $S \text{ member}$  is inductive and  $S, R$  are entire, a pre-fixed point of  $X \mapsto SF(X)R$  is entire

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = g F(X) f \implies \text{Map}(X)$$

(6.5e)

when  $g \text{ member}$  is inductive and  $f, g$  are maps, the solution of  $X = gF(X)f$  is a map

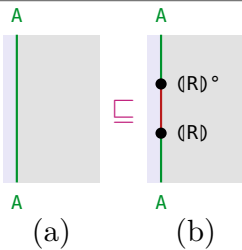


|     |                          |                                |
|-----|--------------------------|--------------------------------|
| (a) | $(g F(Y) f)^\circ X$     | $X = gF(X)f, g \text{ simple}$ |
| (b) | $f^\circ F(Y^\circ X) f$ | $Y^\circ X \sqsubseteq 1$      |
| (c) | $f^\circ f$              | $f \text{ simple}$             |
| (d) | $1$                      |                                |

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \implies 1 \sqsubseteq (R)^\circ (R)$$

(6.5f)

when  $R^\circ \text{ member}$  is inductive and  $R$  is surjective, the fold  $(R)$  is surjective

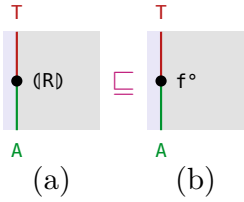


|     |                 |                         |
|-----|-----------------|-------------------------|
| (a) | $1$             | (6.5d) at $S = R^\circ$ |
| (b) | $(R)^\circ (R)$ |                         |

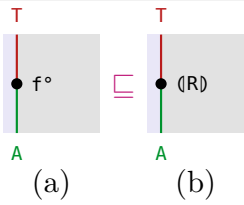
$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \implies f^\circ = \langle R \rangle$$

(6.5g)

a map  $f$  with  $Rf \sqsubseteq F(f)\alpha$ , for a surjective  $R$ , has the fold  $\langle R \rangle$  as its converse



|     |                     |                                                    |
|-----|---------------------|----------------------------------------------------|
| (a) | $\langle R \rangle$ | shunting $f$ , fusion, $Rf \sqsubseteq F(f)\alpha$ |
| (b) | $f^\circ$           |                                                    |

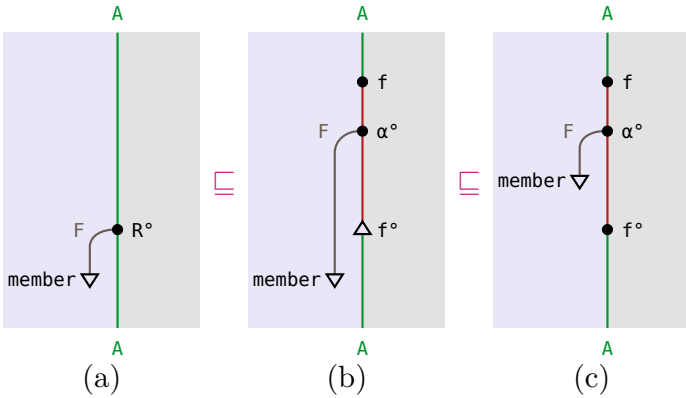


|     |                     |                                                |
|-----|---------------------|------------------------------------------------|
| (a) | $f^\circ$           | (6.5f) and $\langle R \rangle f \sqsubseteq 1$ |
| (b) | $\langle R \rangle$ |                                                |

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \implies R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ$$

(6.5h)

$R^\circ \text{ member}$  is below  $\alpha^\circ \text{ member}$  conjugated by  $f$ , so it is inductive when  $\alpha^\circ \text{ member}$  is



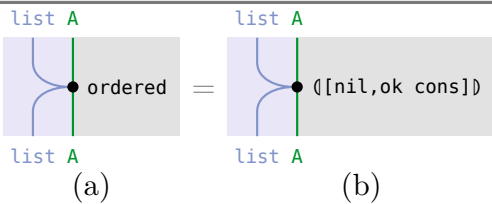
|     |                                            |                                                                                                             |
|-----|--------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| (a) | $R^\circ \text{ member}$                   | $R^\circ \sqsubseteq f \alpha^\circ F(f^\circ) \text{ — } R f \sqsubseteq F(f) \alpha$ , shunting, converse |
| (b) | $f \alpha^\circ F(f^\circ) \text{ member}$ | $\text{member}$ lax natural                                                                                 |
| (c) | $f \alpha^\circ \text{ member } f^\circ$   |                                                                                                             |

## §6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

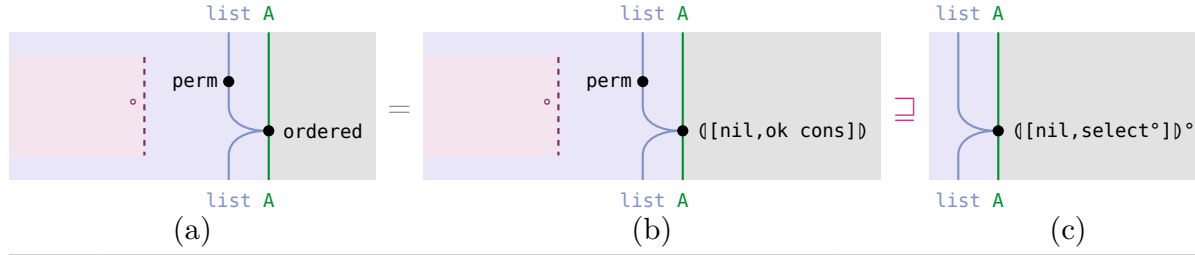


|     |                                                |                 |
|-----|------------------------------------------------|-----------------|
| (a) | $\text{ordered}$                               | fold uniqueness |
| (b) | $\langle [\text{nil}, \text{ok cons}] \rangle$ |                 |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered} \quad (6.6b)$$

every output of unfolding the input by **select** is a sorted permutation of it

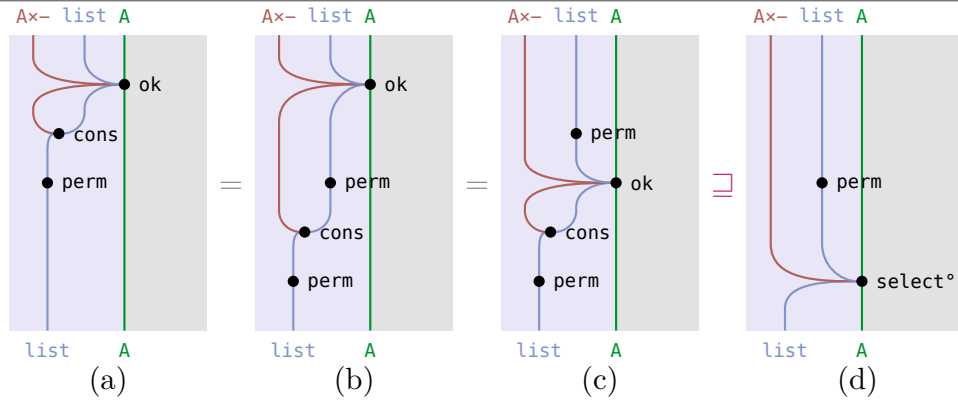
$\text{perm}^\circ = \text{perm}$ ,  $\text{ordered}^\circ = \text{ordered}$



|     |                                                            |                           |
|-----|------------------------------------------------------------|---------------------------|
| (a) | $(\text{ordered perm})^\circ$                              | (6.6a)                    |
| (b) | $(([\text{nil}, \text{ok cons}])^\circ \text{perm})^\circ$ | fusion (6.4) under (6.6c) |
| (c) | $([\text{nil}, \text{select}^\circ])^\circ$                |                           |

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

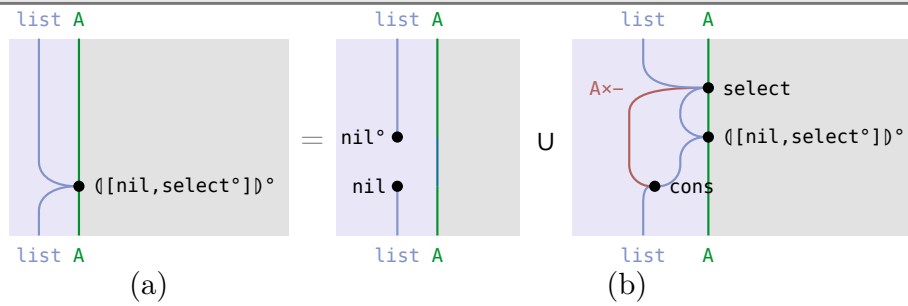
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



|     |                                                      |                                                                                 |
|-----|------------------------------------------------------|---------------------------------------------------------------------------------|
| (a) | $\text{ok cons perm}$                                | $\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$                    |
| (b) | $\text{ok } (1 \times \text{perm}) \text{cons perm}$ | Ex 6.22, $\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$ |
| (c) | $(1 \times \text{perm}) \text{ok cons perm}$         | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$                           |
| (d) | $(1 \times \text{perm}) \text{select}^\circ$         |                                                                                 |

$$([\text{nil}, \text{select}^\circ])^\circ = \text{nil}^\circ \text{nil} \cup \text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on

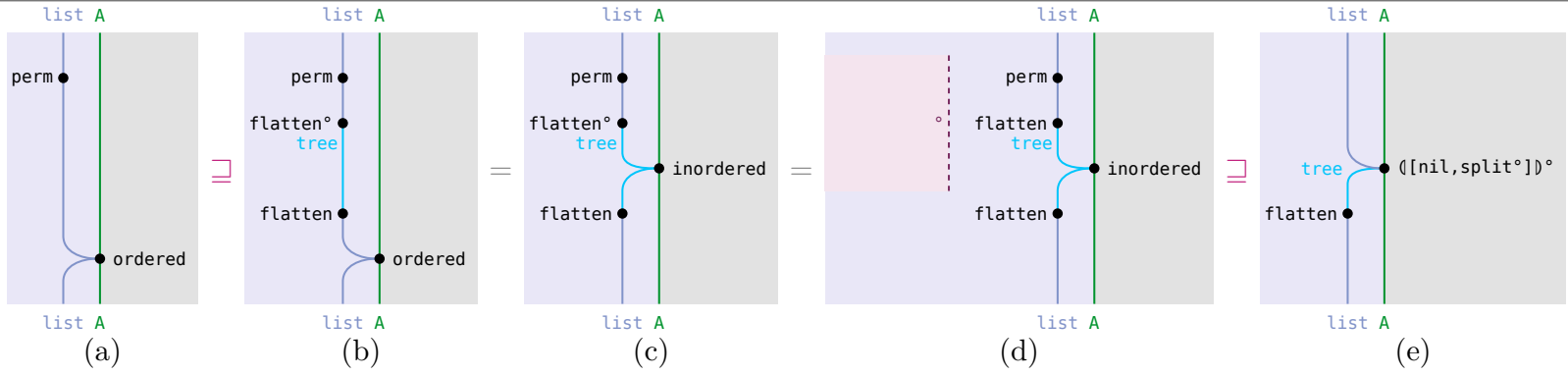


|     |                                                                                                                    |                                  |
|-----|--------------------------------------------------------------------------------------------------------------------|----------------------------------|
| (a) | $([\text{nil}, \text{select}^\circ])^\circ$                                                                        | unfolding the converse of a fold |
| (b) | $\text{nil}^\circ \text{nil} \cup \text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{cons}$ |                                  |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6e)$$

**flatten  $\sqsubseteq$  perm ordered**

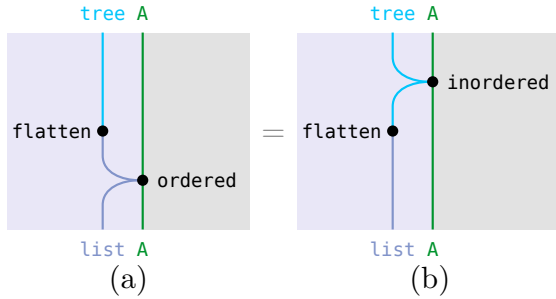
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



|     |                                                            |                                   |
|-----|------------------------------------------------------------|-----------------------------------|
| (a) | perm ordered                                               | flatten is simple                 |
| (b) | perm flatten° flatten ordered                              | flatten ordered=inordered flatten |
| (c) | perm flatten° inordered flatten                            | converses                         |
| (d) | (inordered flatten perm)° flatten                          | fusion (6.4) under (6.6j)         |
| (e) | $([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$ |                                   |

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

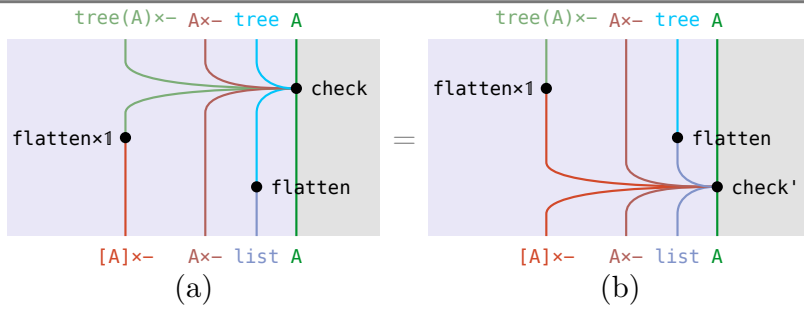
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



|     |                   |                                                                       |
|-----|-------------------|-----------------------------------------------------------------------|
| (a) | flatten ordered   | R transitive: a flattened tree is ordered iff every node passes check |
| (b) | inordered flatten |                                                                       |

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \quad (6.6g)$$

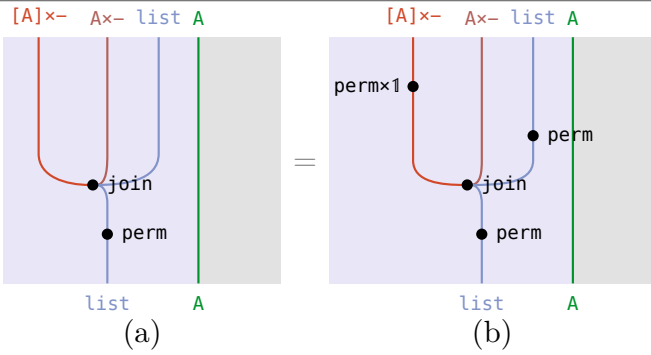
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



|     |                                  |                                       |
|-----|----------------------------------|---------------------------------------|
| (a) | check (flatten × (1 × flatten))  | b intree x $\iff$ b inlist flatten(x) |
| (b) | (flatten × (1 × flatten)) check' |                                       |

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6h)$$

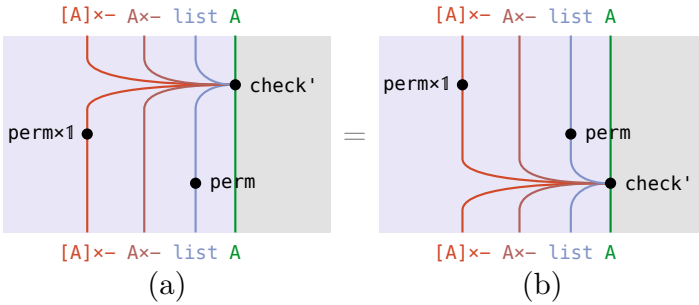
permuting the list  $x \mathrel{++} [a] \mathrel{++} y$  is the same as first permuting  $x$  and  $y$  and then permuting the joined list



|     |                                                                 |                                                   |
|-----|-----------------------------------------------------------------|---------------------------------------------------|
| (a) | $\text{join perm}$                                              | $\text{perm}$ is transitive and closed under $++$ |
| (b) | $(\text{perm} \times (1 \times \text{perm})) \text{ join perm}$ |                                                   |

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{ check}' \quad (6.6i)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

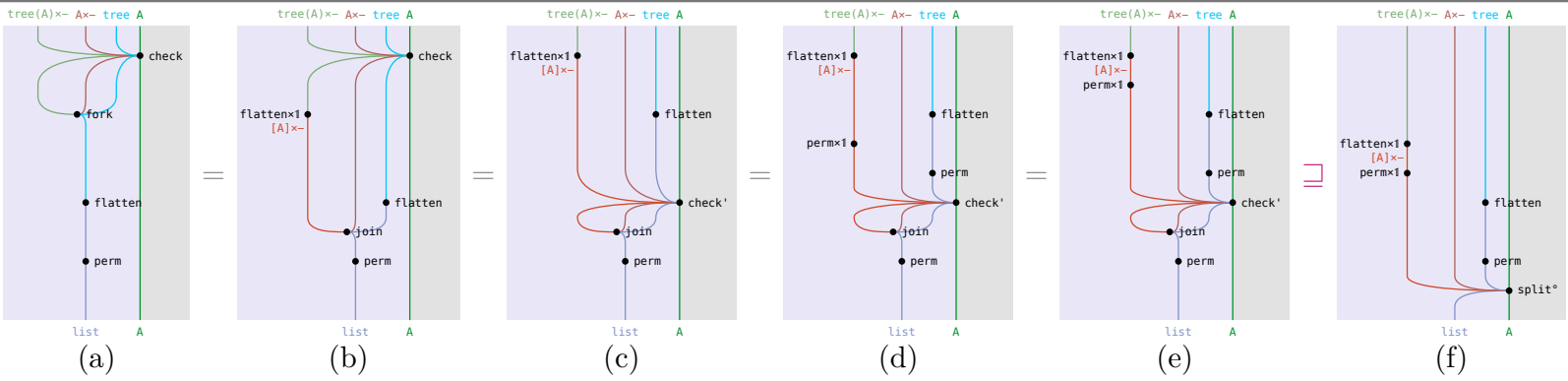


|     |                                                              |                                                             |
|-----|--------------------------------------------------------------|-------------------------------------------------------------|
| (a) | $\text{check}' (\text{perm} \times (1 \times \text{perm}))$  | $b \text{ inlist } x \iff b \text{ inlist } \text{perm}(x)$ |
| (b) | $(\text{perm} \times (1 \times \text{perm})) \text{ check}'$ |                                                             |

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ \sqsubseteq \text{check fork} \quad (6.6j)$$

**flatten perm**

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



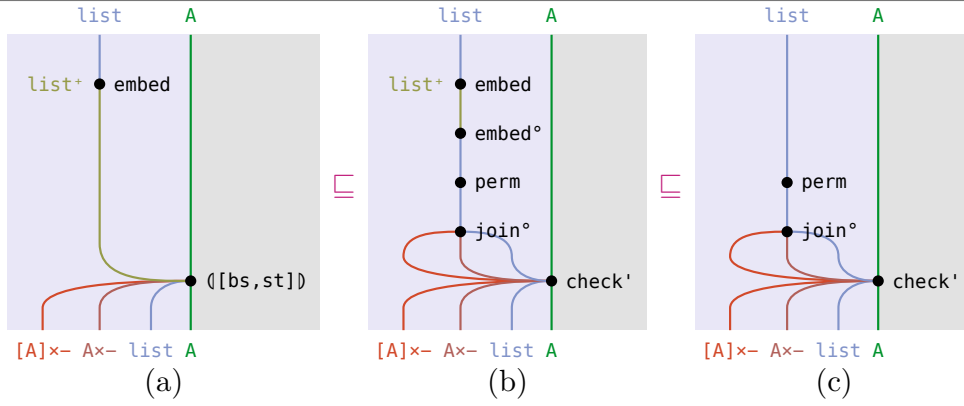
|     |                                                                                                                                  |                                                                                                                   |
|-----|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{check fork flatten perm}$                                                                                                 | $\text{fork flatten} = F(\text{flatten}) \text{ join}$                                                            |
| (b) | $\text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{ join perm}$                                               | $\text{check } F(\text{flatten}) = F(\text{flatten}) \text{ check}'$                                              |
| (c) | $(\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \text{ join perm}$                                             | $\text{join perm} = F(\text{perm}) \text{ join perm, check'}$<br>$F(\text{perm}) = F(\text{perm}) \text{ check}'$ |
| (d) | $(\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{ check}' \text{ join perm}$ | functors                                                                                                          |
| (e) | $((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ check}' \text{ join perm}$                               | $\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$                                                  |
| (f) | $((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{ split}^\circ$                                            |                                                                                                                   |

**bs**  $\sqsubseteq$  wrap perm join $^\circ$  check'  $\wedge$  ( $1 \times$  (perm join $^\circ$  check')) st  $\sqsubseteq$  cons perm join $^\circ$  check'  $\Rightarrow$  embed (6.6k)

$\llbracket \text{bs}, \text{st} \rrbracket \sqsubseteq$  perm join $^\circ$  check'

turning a list into a non-empty list and folding it with any  $[\text{bs}, \text{st}]$  meeting the two conditions below gives only

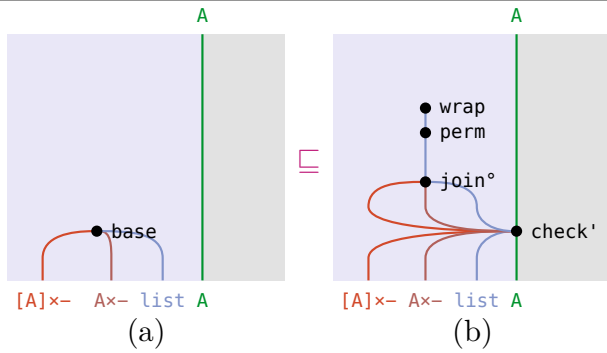
**checked triples**  $(x, a, y)$  whose join  $x \# [a] \# y$  is a permutation of the list



|     |                                                    |                                                    |
|-----|----------------------------------------------------|----------------------------------------------------|
| (a) | embed $\llbracket \text{bs}, \text{st} \rrbracket$ | (6.6l) and (6.6m), induction on the non-empty list |
| (b) | embed embed $^\circ$ perm join $^\circ$ check'     | embed is simple                                    |
| (c) | perm join $^\circ$ check'                          |                                                    |

**base**  $\sqsubseteq$  wrap perm join $^\circ$  check' (6.6l)

**base(a)** is a **checked triple** whose join is a permutation of the one-element list  $[a]$

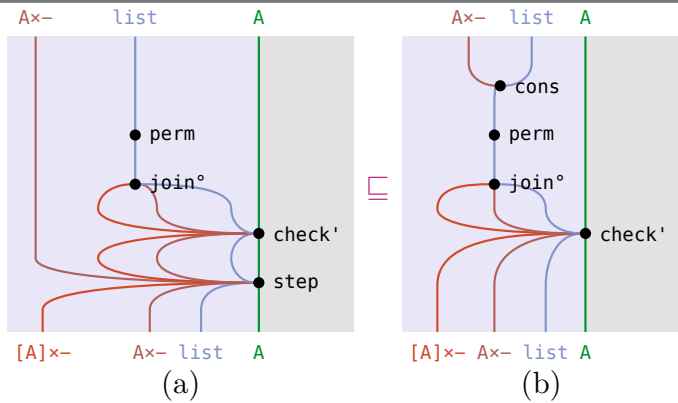


|     |                                |                                                           |
|-----|--------------------------------|-----------------------------------------------------------|
| (a) | base                           | $[] \# [a] \# [] = [a]$ , and check' holds on empty lists |
| (b) | wrap perm join $^\circ$ check' |                                                           |

$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}'))$  (6.6m)

**step**  $\sqsubseteq$  cons perm join $^\circ$  check'

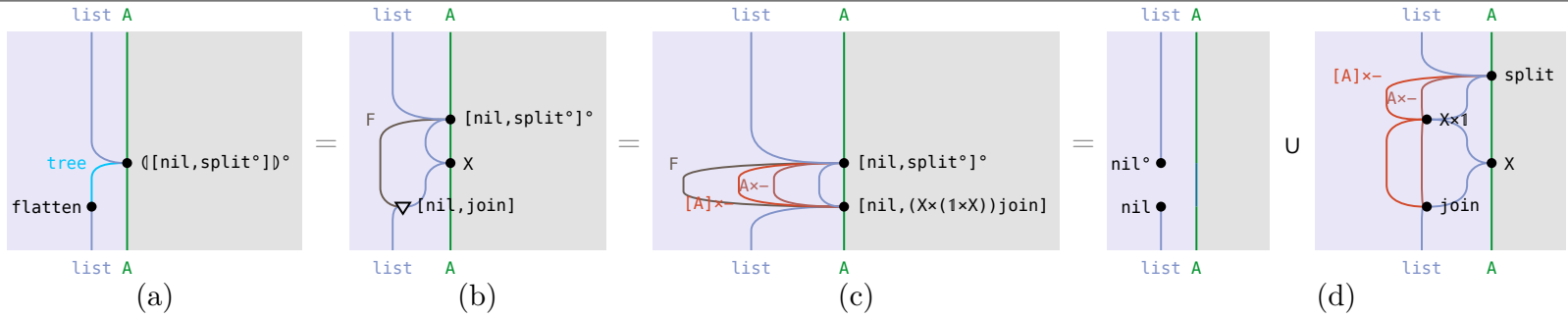
if  $(x, b, y)$  is a **checked triple** for a permutation of  $\mathbf{l}$ , then **step(a, (x, b, y))** is a **checked triple** for a permutation of  $[a] \# \mathbf{l}$



|     |                                                          |                                                                               |
|-----|----------------------------------------------------------|-------------------------------------------------------------------------------|
| (a) | $(1 \times (\text{perm join}^\circ \text{check}'))$ step | $aRb$ puts $a$ in front of $x$ ; otherwise $bRa$ and $a$ goes in front of $y$ |
| (b) | cons perm join $^\circ$ check'                           |                                                                               |

$$([nil, split]^{\circ})^{\circ} \text{ flatten} = nil^{\circ} nil \cup split (([nil, split]^{\circ})^{\circ} \text{ flatten}) \times (1 \times ([nil, split]^{\circ})^{\circ} \text{ flatten})) \text{ join} \quad (6.6n)$$

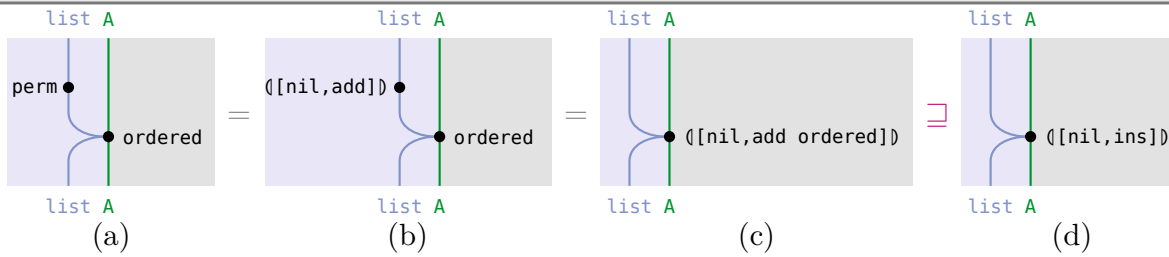
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



|     |                                                            |                                                        |
|-----|------------------------------------------------------------|--------------------------------------------------------|
| (a) | $([nil, split]^{\circ})^{\circ} \text{ flatten}$           | hylomorphism theorem                                   |
| (b) | $[nil, split]^{\circ} F(X) [nil, join]$                    | $F(X) [nil, join] = [nil, (X \times 1 \times X) join]$ |
| (c) | $[nil, split]^{\circ} [nil, (X \times (1 \times X)) join]$ | coproduct                                              |
| (d) | $nil^{\circ} nil \cup split (X \times (1 \times X)) join$  |                                                        |

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([nil, ins]) \sqsubseteq \text{perm ordered} \quad (6.6o)$$

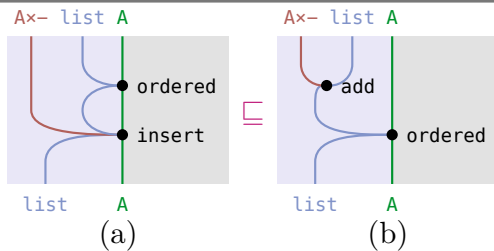
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



|     |                        |                                                                        |
|-----|------------------------|------------------------------------------------------------------------|
| (a) | perm ordered           | perm = $([nil, add])$                                                  |
| (b) | $([nil, add])$ ordered | fusion, add ordered = $(1 \times \text{ordered})$ add ordered          |
| (c) | $([nil, add ordered])$ | $(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$ |
| (d) | $([nil, ins])$         |                                                                        |

$$(\forall a. \forall b. \text{leb}(a,b) = \text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b) = \text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered} \quad (6.6p)$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



|     |                                            |                                                               |
|-----|--------------------------------------------|---------------------------------------------------------------|
| (a) | $(1 \times \text{ordered}) \text{ insert}$ | insert splices <b>a</b> in, and <b>R</b> transitive and total |
| (b) | add ordered                                |                                                               |

## §6.7. Closure

$$R^* = (\mu X : 1 \cup X R) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R\***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R\***

$$1 \sqsubseteq R^* \quad (6.7c)$$

**R\*** is reflexive

|               |     |                |                      |
|---------------|-----|----------------|----------------------|
|               | (a) | 1              |                      |
| $\sqsubseteq$ | (b) | $1 \cup R R^*$ | union                |
| $=$           | (c) | $R^*$          | fixed point, Thm 6.1 |

$$R \subseteq R^*$$

(6.7d)

 $R^*$  contains  $R$ 

|               |     |           |                              |
|---------------|-----|-----------|------------------------------|
|               | (a) | $R$       |                              |
| =             | (b) | $R$       | identity                     |
| $\sqsubseteq$ | (c) | $R \ R^*$ | $1 \sqsubseteq R^*$ — (6.7c) |
| $\sqsubseteq$ | (d) | $R^*$     | fixed point                  |

$$R^* \ R^* \subseteq R^*$$

(6.7e)

 $R^*$  is transitive, because  $R^*/R^*$  is a prefixed point of  $X \mapsto 1 \cup R X$ 

|               |     |                                |                                                                     |
|---------------|-----|--------------------------------|---------------------------------------------------------------------|
|               | (a) | $(1 \cup R (R^* / R^*)) \ R^*$ |                                                                     |
| =             | (b) | $R^* \cup R (R^* / R^*) \ R^*$ | $(P \cup Q) T = P T \cup Q T$ — composition distributes over $\cup$ |
| $\sqsubseteq$ | (c) | $R^* \cup R \ R^*$             | $(S/S) S \subseteq S$                                               |
| $\sqsubseteq$ | (d) | $R^*$                          | $R R^* \subseteq R^*$ — (6.7d)                                      |

$$1 \in X \wedge X X \subseteq X \wedge R \subseteq X \implies R^* \subseteq X$$

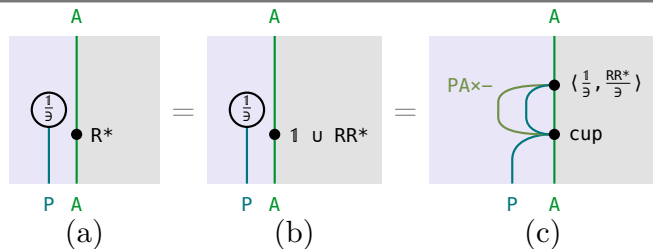
(6.7f)

a reflexive transitive  $X$  containing  $R$  is a prefixed point of  $X \mapsto 1 \cup R X$ , so it contains  $R^*$ 

|               |     |              |                          |
|---------------|-----|--------------|--------------------------|
|               | (a) | $1 \cup R X$ |                          |
| $\sqsubseteq$ | (b) | $X \cup X X$ | $1 \in X, R \subseteq X$ |
| $\sqsubseteq$ | (c) | $X$          | $X X \subseteq X$        |

$$\frac{R^*}{\exists} = \left( \frac{1}{\exists}, \frac{R \ R^*}{\exists} \right) \text{ cup}$$

(6.7g)

the set of  $R^*$ -successors of  $a$  is  $a$  itself joined with the  $R^*$ -successors of its  $R$ -successors


|     |                                                                         |                                                          |
|-----|-------------------------------------------------------------------------|----------------------------------------------------------|
| (a) | $\frac{R^*}{\exists}$                                                   | $R^* = 1 \cup R R^*$                                     |
| (b) | $\frac{1 \cup R \ R^*}{\exists}$                                        | $\wedge (R \cup S) = (\wedge(R), \wedge(S)) \text{ cup}$ |
| (c) | $\left( \frac{1}{\exists}, \frac{R \ R^*}{\exists} \right) \text{ cup}$ |                                                          |

$$R - \emptyset = R$$

(6.7h)

taking nothing away leaves  $R$ 

$$R \cup (S - R) = R \cup S$$

beside  $R$ , only the part of  $S$  outside  $R$  adds anything

$$R - (S \cup T) = R - S - T$$

taking away  $S$  and then  $T$  takes away  $S \cup T$ 

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking  $T$  away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$$

(6.7i)

the least fixed point of  $\varphi$  after  $\psi$  is  $\varphi$  applied to the least fixed point of  $\psi$  after  $\varphi$ 

$$S \ R^* = (\mu X : S \cup X \ R)$$

(6.7j)

 $S$  followed by any number of  $R$  steps is the least  $X$  containing  $S$  and closed under a further  $R$  step

$$R^* \ S = (\mu X : S \cup R \ X)$$

any number of  $R$  steps followed by  $S$  is the least  $X$  containing  $S$  and closed under an  $R$  step in front

$$\theta(\emptyset, S) = S \ R^*$$

(6.7k)

started with nothing found and  $S$  to explore,  $\theta$  computes  $S R^*$ 

|   |     |                                                       |                              |
|---|-----|-------------------------------------------------------|------------------------------|
|   | (a) | $\theta(\emptyset, S)$                                |                              |
| = | (b) | $\emptyset \cup (\mu X : S \cup (X \ R - \emptyset))$ | definition of $\theta$       |
| = | (c) | $(\mu X : S \cup X \ R)$                              | $R - \emptyset = R$ — (6.7h) |
| = | (d) | $S \ R^*$                                             | (6.7j)                       |

| $\theta(P, \emptyset) = P$ <span style="float: right;">(6.71)</span> |     |                                             |                                                                |
|----------------------------------------------------------------------|-----|---------------------------------------------|----------------------------------------------------------------|
| with nothing left to explore, $\theta$ returns what it has found     |     |                                             |                                                                |
|                                                                      | (a) | $\theta(P, \emptyset)$                      |                                                                |
| =                                                                    | (b) | $P \cup (\mu X : \emptyset \cup (X R - P))$ | definition of $\theta$                                         |
| =                                                                    | (c) | $P \cup \emptyset$                          | $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$ |
| =                                                                    | (d) | $P$                                         | $P \cup \emptyset = P$                                         |